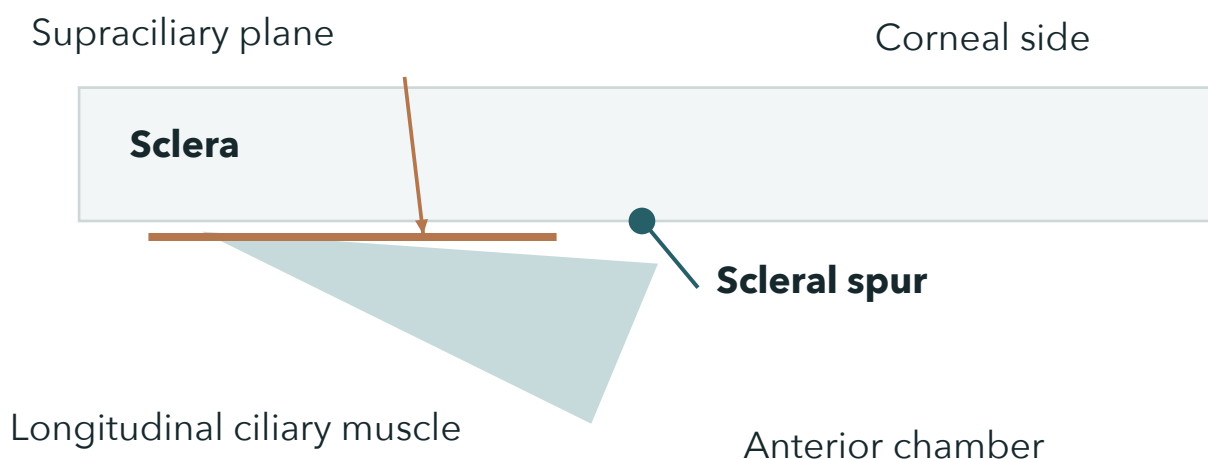


Anterior segment tissue planes

Part 1 | The angle as a tissue plane

The scleral spur is a structural junction, not just a gonioscopic landmark.



A cleft separates muscle attachment from spur

Stylised limbal section shows anterior chamber, scleral spur, ciliary muscle and supraciliary plane.

Connect anatomy to flow

Longitudinal ciliary muscle attaches near the scleral spur. Trabecular drainage and supraciliary drainage occupy different tissue planes. Describe a traumatic gap by the boundaries that have separated, then connect the resulting communication to its pressure effect.

Recession and cyclodialysis differ

Intramuscular injury and separation from sclera have different hydraulic consequences.

Compare the split

Feature	Angle recession	Cyclodialysis
Plane	Between muscle fibre groups	Muscle separates from spur
Drainage opening	No direct new supraciliary channel	Anterior chamber to supraciliary space
Pressure concern	Later secondary glaucoma	Hypotony may dominate

Cleft mechanism

- Loss of ciliary attachment
- ↓
- Abnormal drainage into suprachoroidal space
- ↓
- Reduced pressure and tissue deformation

Do not infer mechanism from IOP alone

Trauma can also reduce secretion by damaging the ciliary body. Demonstrating the structural communication is more specific than labelling every soft eye as cyclodialysis.

Nonpenetrating dissection

Deep sclerectomy deliberately retains a thin filtration barrier.

Reconstruct the operative depth

- Raise a superficial scleral flap
↓
- Remove the deep scleral tissue
↓
- Unroof the external Schlemm canal wall
↓
- Leave the trabeculo-Descemet membrane intact

What remains

Finding	Interpretation
Slow aqueous percolation	A residual membrane provides resistance
Anterior chamber intact	Nonpenetrating plane retained
Membrane perforation	Anterior-chamber entry has occurred

Translate the vocabulary

Nonpenetrating describes the retained interface. It does not mean that the dissection never approaches the anterior chamber or that no aqueous traverses the remaining tissue.

Choose an image for a hidden plane

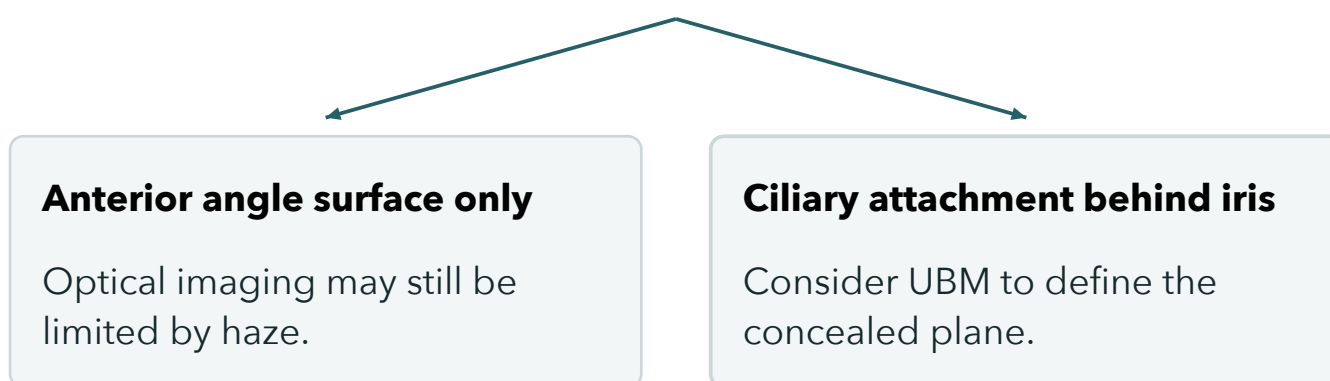
Optical access and acoustic access are different limitations.

Target

Feature	Imaging principle	Anatomical implication
AS-OCT	Optical backscatter	Good anterior angle detail
UBM	High-frequency ultrasound	Can assess tissue behind iris pigment
Gonioscopy	Direct clinical angle view	Needs usable optical view

When the cornea is hazy

What must be demonstrated?



Pressure after a cleft closes

Restoring an attachment can remove a major low-resistance pathway.

Sequential inference

- Cleft permits abnormal drainage
↓
- Repair apposes separated tissues
↓
- Abnormal route is closed
↓
- Pressure can rise as normal resistance returns

The useful comparison

Hypotony before repair does not prove healthy trabecular function.

Traumatic recession can coexist with cyclodialysis.

Post-repair pressure reflects secretion, conventional resistance and downstream pressure.

Examination approach

Choose the mechanism supported by the changed anatomy. Do not assume that a pressure rise after successful closure means the repair failed; closure itself alters the hydraulic system.